ChE 312 – Transport Processes II

0. Course introduction, continued.

Motivate the study of mass transfer by examining distillation and membrane separations.

I. Distillation: The dominant large-scale separation process – presently.

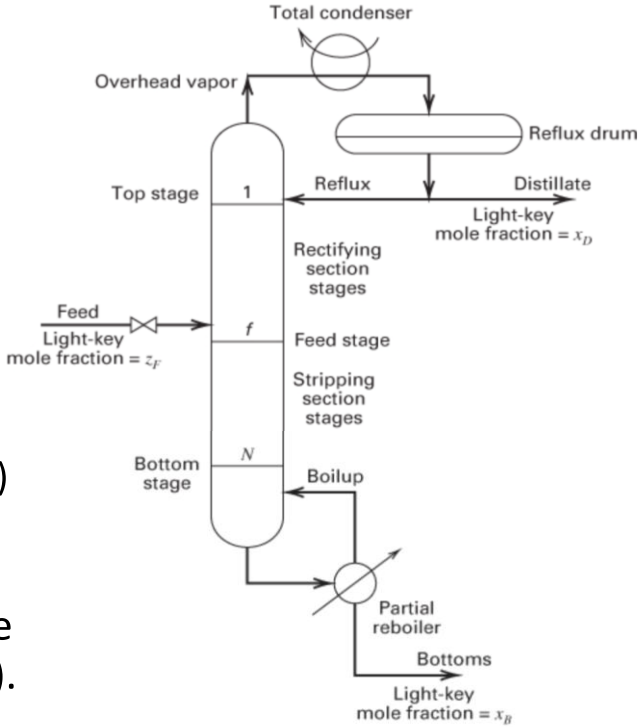
We can quickly learn how to do in-depth analysis of stagewise equilibrium separation.

Solve mass balances with equilibrium relations.

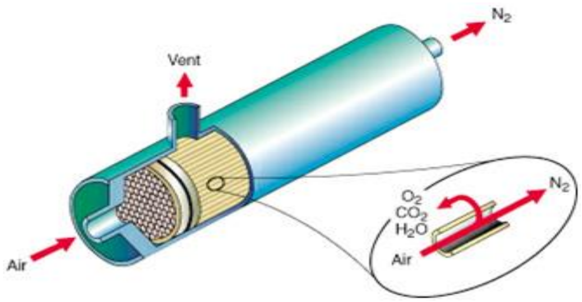
Model a simple still.

Continuous distillation: The biggest thermally driven separation.

- Crude oil has components ranging from dissolved gases to light liquids to heavy liquids to asphalt, so hight temperatures are required.
  - Represents about 8% of US energy use!
  - Heat is generated by combustion in huge furnaces -- generates lots of CO_2.
- Could we keep the process but not use combustion?
  - Maybe use electricity (from what?).
  - However, delivering such large amounts of heating (and cooling) from electricity isn't yet practical.
- Let’s look at Houston, starting with ExxonMobil's chemical plant and refinery in Baytown, Texas and zooming out to see other facilities on the Houston Ship Channel (Buffalo Bayou).



Compare to membrane separations.



- View the make-up of a shell-and-tube membrane separator.
- It's not driven thermally, which saves huge amounts of energy and CO_2 generation.
- However, we can't do much to analyze it quantitatively without a way to model mass transfer through the membrane.
- We'll eventually analyze and design it using the same types of series-resistance overall transfer coefficients as we used in heat exchangers, incorporating convective mass-transfer coefficients and diffusivity terms.

What drives separations -- and mixing? Approach toward equilibrium.

- How do they occur? Mass transfer!

- What differentiates separation processes? One thing is the type of driving force, like Δ(concentration - equilibrium concentration).

- How do you cause a separation?
  - Phase creation, as in distillation or in chromatography, where you adsorb molecules from a gas or liquid onto and off a column surface.
  - Phase addition, as in liquid-liquid extraction;
  - Gradient across a barrier, as in membrane ultrafiltration;
  - Application of an external field, as in centrifugation or electrophoresis.

Last time: Obtaining quantitative equilibrium relations is crucial to all mass transfer processes.

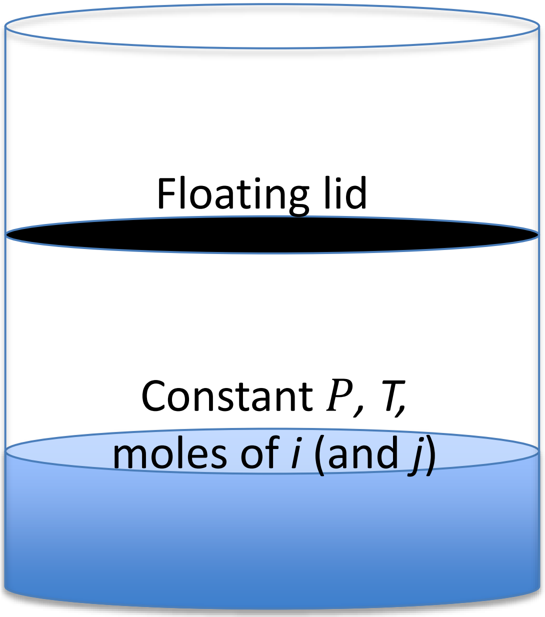
- Start with phase equilibrium for ideal-gas mixtures and ideal solutions; that is, Raoult's "Law” $y_i P = x_i P_{vap,i}$
- "Volatility" y_i/x_i and ”relative volatility" $\alpha_{i,ref} = \frac{y_i/x_i}{y_{ref}/x_{ref}}$ are useful, especially for binary VLE when relative volatility is almost constant, giving $y = \frac{\alpha x}{1+x(\alpha-1)}$
- If a mixture is nonideal, *y-vs-x* relations still must approach equilibrium limits, so approximate $y(x)$ by adding a nonlinear term: $y = \frac{\alpha x}{1+x(\alpha-1)} + b(x)(1-x)$
- For careful calculations, use Raoult's Law with fugacity coefficients and activity coefficients: $\phi_i y_i P = \gamma_i x_i P_{vap,i}$

Last time, we considered “static” VLE and Raoult’s “Law.”

Vapor-Liquid Equilibrium (VLE): A pure species will exist as either liquid or gas, and the transition is at $P = P_{vap,i}(T)$.

It seems reasonable that mixed ideal gases would contribute to total pressure based on fractional number of moles: $P = \sum y_i P$

It seems reasonable that an ideal solution would contribute to total pressure in the same way: $P = \sum x_i P_{vap,i}(T)$



Raoult’s Law combines these idealizations:

$$y_i P = x_i \cdot P_{vap,i}(T)$$

where we correlate $P_{vap,i}(T)$ data with Antoine equations

$$log\left(P_{vap,i}\right) = A_i - \frac{B_i}{C_i + T}$$

Raoult’s “Law” is flawed, especially for liquids.

- To satisfy the idealizations, molecules wouldn’t interact at all.
  - That’s not the truth, but in many useful cases, it is close enough.
- Throughout engineering, we chose idealizations and apply correction factors:
  - For nonideal gas *P*-*V*-*T* behaviors, apply a compressibility factor *Z*, so $PV = ZnRT$
  - For phase equilibrium, correct gas nonideality and liquid nonideality separately: $y_i P = x_i P_{vap,i}$ becomes $\phi_i y_i P = \gamma_i x_i P_{vap,i}$
  - The “fugacity coefficient” ϕ_i depends on y_i, P and T, and it can be computed from an equation of state; it is often near 1.0.
  - The “activity coefficient” γ_i depends on x_i and T and the chemical species involved, and often $\gamma_i \neq 1$ <u>except</u> always $\gamma_i \to 1$ when species i approaches purity.

We often have the x_is and want to have y_is, or vice versa.

- If $\phi_i \cong 1$ and $\gamma_i \cong 1$, you can solve multicomponent ideal VLE.
  - In general, $\phi_i y_i P = \gamma_i x_i P_{vap,i}$ so $\sum(\phi_i y_i P) = \sum(\gamma_i x_i P_{vap,i})$
  - Which becomes $P = \sum[x_i \cdot P_{vap,i}\ (T)]$
- If you have Antoine coefficients for all the $P_{vap,i}(T)$, then:
  - With T and the x_is, you can get P at equilibrium
  - or with P and the x_is, you can get T at equilibrium.
  - Then use $y_i \cdot P = x_i \cdot P_{vap,i}\ (T)$ to get y_i from x_i, or vice versa.
- If $\phi_i \cong 1$ and you have equations for all the γ_i, you can solve multicomponent nonideal VLE.

Write as a solvable algorithm if you know P and x_A.

$$y_A P = x_A P_{vap,A}(T)$$
where $\log_{10} P_{vap,A}(T) = \mathcal{A}_A - \frac{\mathcal{B}_A}{T+\mathcal{C}_A}$

$$y_B P = (1 - x_A) P_{vap,B}(T)$$
where $\log_{10} P_{vap,B}(T) = \mathcal{A}_B - \frac{\mathcal{B}_B}{T+\mathcal{C}_B}$

$$y_A P + y_B P = x_A P_{vap,A}(T) + (1 - x_A) P_{vap,B}(T) = P$$
gives
$$x_A P_{vap,A}(T) + (1 - x_A) P_{vap,B}(T) - P = 0$$

For P and x_A, solve for T numerically using a root finder like Excel Solver;
Then calculate $y_A = \frac{x_A P_A(T)}{P}$ using T and x_A.

Another approach is K value: Volatility or "partition coefficient."

- $K_i = (y_i/x_i)_{eq}$; the volatility for vapor-liquid equilibrium (VLE).
  - Can find K_i for any phase equilibria; i.e., $K_i = \left(x_{i,phase\ 1}/x_{i,phase\ 2}\right)_{eq}$
  - For liquid-solid, adsorbate-surface, liquid-liquid: $\gamma_{i,I} x_{i,I} \cancel{P_{vap,i}} = \gamma_{i,II} x_{i,II} \cancel{P_{vap,i}}$ so $K_i = \left(\gamma_{i,II}/\gamma_{i,I}\right)_{eq}$
  - "Oil-water partition coefficient" $K_{ow} = \left(x_{i,oil-rich\ phase}/x_{i,water-rich\ phase}\right)_{eq}$ is an important indicator for bioaccumulation.
  - Various correlations; increasingly predictable by molecular modeling.

Relative volatility α_{AB} compares volatility of A vs B.

- Form useful equations:

$$\boxed{\alpha_{AB}} \equiv \frac{y_A / x_A}{y_B / x_B} \xrightarrow{\text{For a binary}} \frac{y_A / x_A}{(1 - y_A)/(1 - x_A)} \xrightarrow{\text{If ideal (Raoult's)}} \frac{P_{vap,A} / P}{P_{vap,B} / P} \boxed{= \frac{P_{vap,A}}{P_{vap,B}}}$$

- For ideal binary mixture, α_{AB} is a function only of T; sometimes it is ~constant.
- Define y as y_A, where A is the more volatile component of the two;
- Solve to get $y = f(x_A, T) = \frac{\alpha x}{1+x(\alpha-1)}$ or $x = \frac{y}{\alpha-y(1-\alpha)}$
- Deviations are worse for liquids (x) and are worst for x in the mid-range.
- One useful fitting form is then: $y = \frac{ax}{1+x(a-1)} + b(x)(1-x)$ [note: a isn't α].

Look briefly at Aspen PLUS software for phase equilibrium; we will later use it for design.

- Commercial software packages for bio/chemical process simulation, design, and analysis have made detailed design, analysis, and costing much easier.
- ASPEN Plus is one of several such packages, and we have access to it here. Three other widely used packages are gPROMS, ChemCAD, and ProSim.
- Today we'll look at how to use Aspen Plus's equilibrium-properties tools.
  - Flowsheets can also be constructed for various separation processes with modules such as Radfrac, which models distillation columns to incorporate pressure drop through the column, nonideal equilibria, and more.
- Set up systems and find equilibrium relations: Txy, Pxy, y-vs-x, activity coefficients.

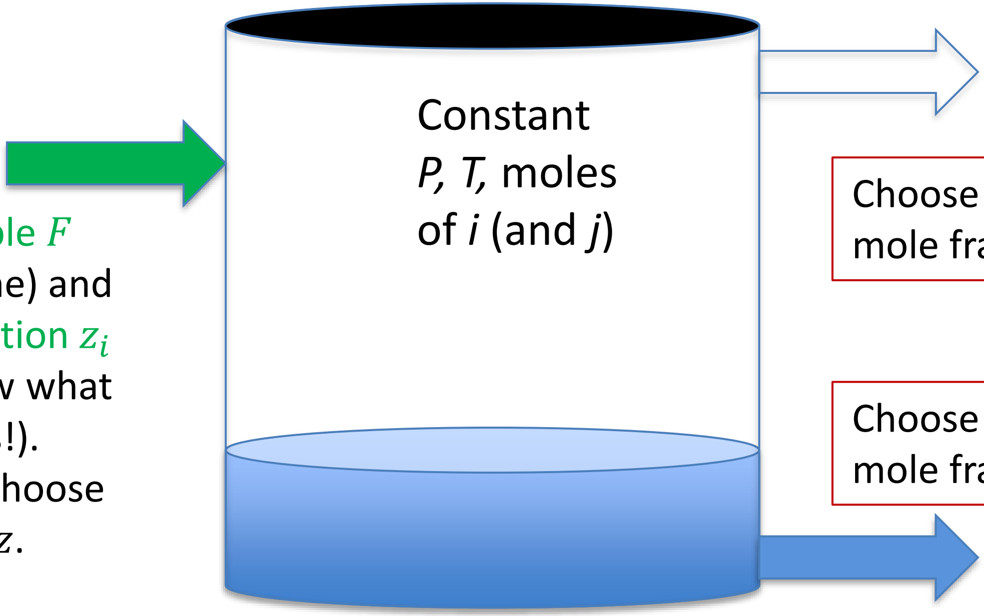
General, basic approaches to analysis and design:

- ******Four essential steps for analysis, design, and creation******
  Sketch and label:
  - Use this info to consider what your objective is and what you know.
- **Write balances ("conservation equations"): Accumulation = In - Out + Generation - Depletion**
  - Macroscopic or differential balance; Mass, mole, energy amounts or rates. Stagewise, continuous, or time-dependent?
- **Obtain properties ("constitutive relations"):**
  - Equilibrium relations: Start with ideal-gas mixtures and ideal solutions (Raoult's "Law"), move to nonideal VLE, multicomponent, and quantifying mass-transfer limitations.
- **Solve and assess results for plausibility:**
  - First decide how to solve. Algebraically solve precise balances and equilibria or
  - integrate differential equations with initial and boundary conditions?

- Graphs vs. analytical integration vs. one-off computer programs vs. standard apps (like Aspen Plus).
- Write overall and localized equations: Mass or mole balances and VLE equations.

Let’s start with a currently and historically important process: DIstillation.

Build on the continuous “flash drum” case:



Choose flowrate variable V, mole fraction of *i* as y_i or y

Choose flowrate variable L and mole fraction of *i* as x_i or x

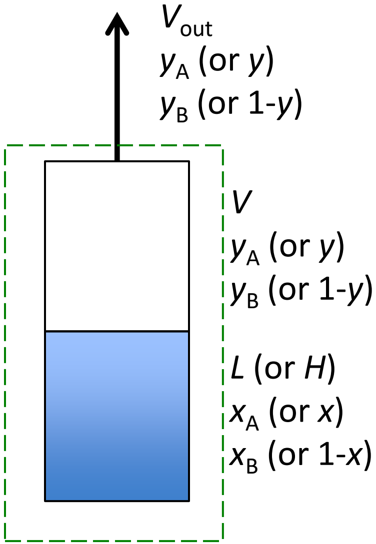
Choose flowrate variable F (moles or mass per time) and feed mass or mole fraction z_i because we don’t know what the phase is (or phases!). If binary, you can just choose one species and write z.

Write two conservation equations: $F = V + L$ and $zF = yV + xL$

And co-solve with equilibrium relation(s), such as $y = \alpha_{ij}x/\left[1 + \left(\alpha_{ij} - 1\right)x\right]$

For "semi-batch" still, set up material balances.

- Batch of liquid, but vapor leaves.
- Choose system boundaries:
  - Just around liquid?
  - Just gas?
  - Overall?
- Define variables:
  - Moles in each phase
  - Mole fractions
  - The exit flowrate
- Not steady state, but if done slowly enough, maintain equilibrium.



Set up material balances.

- Write balances and solve:

$$\text{Accum} = \text{In} - \text{Out} + \text{Gen} - \text{Depl}$$

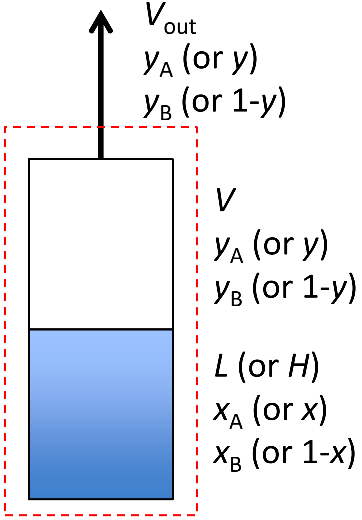
$$\frac{dL}{dt} = 0 - V_{out} + 0 - 0$$

$$\frac{d(xL)}{dt} = 0 - yV_{out} + 0 - 0$$

$$\text{Solving, } \frac{d(xL)}{dt} = x\frac{dL}{dt} + L\frac{dx}{dt} = -y\left(-\frac{dL}{dt}\right)$$

$$\text{Collecting terms, } xdL + Ldx = ydL$$

$$\frac{dL}{L} = \frac{dx}{y - x}$$



- To solve, need to have $y_{eq} = f(x)$

To get numbers, you need to have y as an analytic function $y_{eq} = f(x)$ or as a tabulation or graph.

- Then you have a relation you can integrate analytically or numerically:

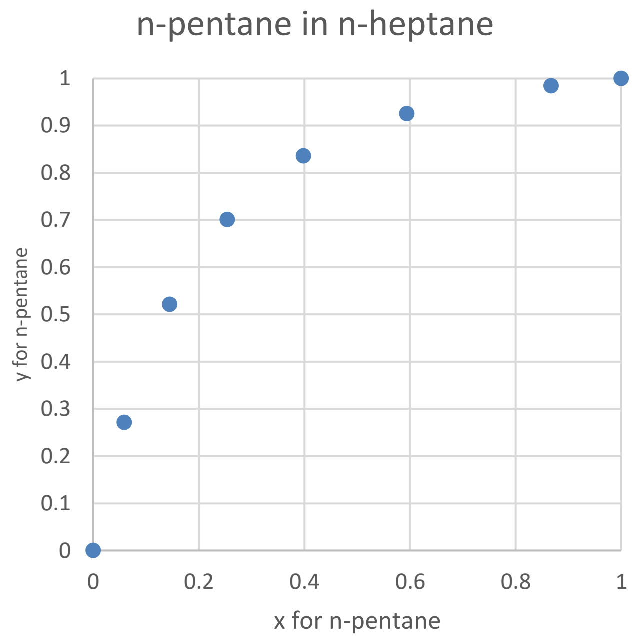
$$\frac{dL}{L} = \frac{dx}{f(x) - x}$$

- Need boundary conditions like initial L_o and x_o.
- Integrate to get $x = g(L)$, noting that time and T are not explicit,
- Or x at a final L --- and thus $y_{eq} = f(x)$, too,
- Or L for a final desired x or y_{eq}.

Example 26.3-2. Simple differential distillation.

- 100 mol initially, 50.0 mol % *n*-pentane mixture with *n*-heptane (should be ideal)
- Differential distillation at P=101.3 kPa until 40 mol are distilled.
- Average composition of total vapor and remaining liquid?

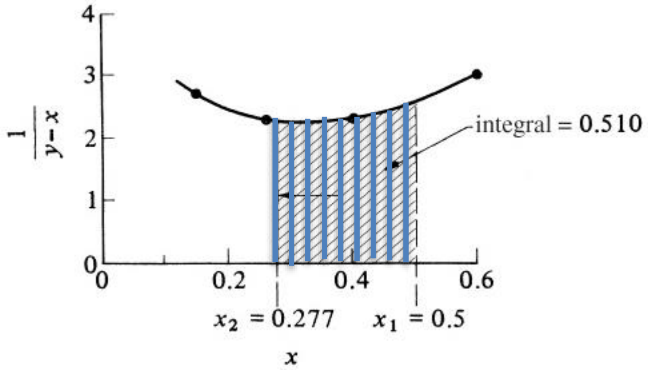
x_{eq}, *n*-pentane in *n*-heptane	y_{eq}, *n*-pentane in *n*-heptane
1.000	1.000
0.867	0.984
0.594	0.925
0.398	0.836
0.254	0.701
0.145	0.521
0.059	0.271
0.000	0.000



Solve using $\frac{dL}{L} = d \ln L = \frac{dx}{y_{eq} - x}$

- Integrate from $L_1 = 100\ mol$ and $x_1 = 0.500$ to $L_2 = 60$ using $ln \frac{L_2}{L_1} = \int_{x_1}^{x_2} \left(\frac{1}{y_{eq} - x} \right) dx$, the Rayleigh equation.
- Then $\ln \frac{L_2}{L_1} = \frac{60}{100} = 0.510.$
- Before-and-after mole balance is: $L_1 x_1 = L_2 x_2 + (L_1 - L_2) y_{ave}$, giving $y_{ave} = \frac{L_1 x_1 - L_2 x_2}{L_1 - L_2}$
- Thus, we need to find x_2 from the integration.

- One approach: Integrate using a graph+table of $\frac{1}{y_{eq} - x}$ vs. x as a sum of $\left(\frac{1}{y_{eq}(x) - x} \right)_{i,ave} \cdot (x_{i+1} - x_i)$



Then x_2 =0.277 gives y_{ave} =0.835.

Solve using $\frac{dL}{L} = d\ \ln L = \frac{dx}{y_{eq}-x}$

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- Thus, we need to find x_2 from the integration.

- Or fit $y = \frac{\alpha x}{1+x(\alpha-1)}$

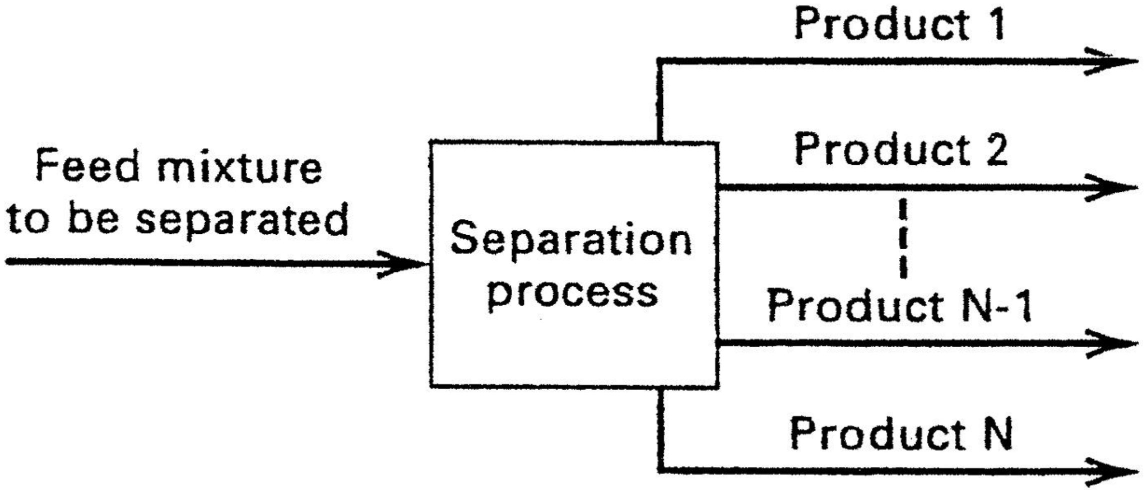
αx/(1+x(α-1))	$(y_{Eq}-y_{Pred})^2$
1.0000	0.0000
0.9774	0.0000
0.9065	0.0003
0.8142	0.0005
0.6929	0.0001
0.5292	0.0001
0.2936	0.0005
0.0000	0.0000
	0.0015
alpha = 6.63	

to get α and compute the integral:

$$\sum\left(\frac{1}{\frac{\alpha x}{1+x(\alpha-1)}-x}\right)_{i,ave}\cdot(x_{i+1}-x_i)$$

using a set of x values.

Now consider multi-tray, continuous distillation.



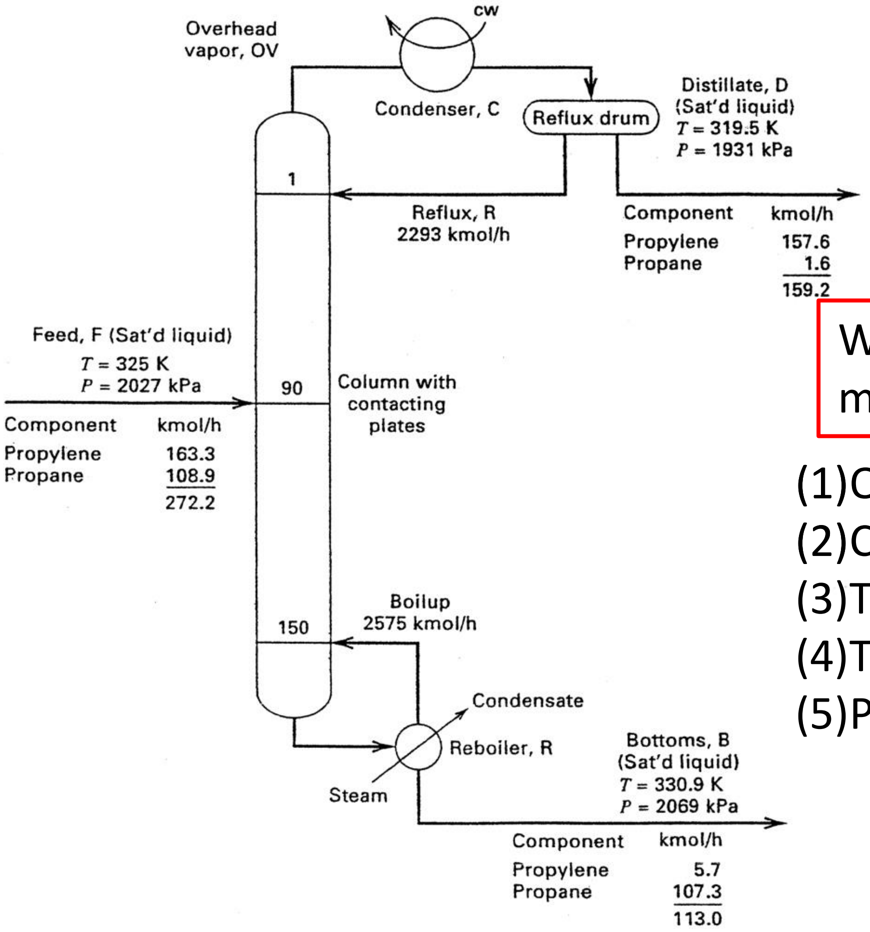
Study https://www.youtube.com/watch?v=I70jgRpf80o (auf Deutsch)

Fig. 1.5; Seader, Henley, and Roper, *Separation Process Principles,* 3rd Ed.

Distillation is a workhorse for separations.

What makes it happen?

Supply heat thru feed and reboiler; Supply coolant at top to return liquid; Then countercurrent L-V contacting.

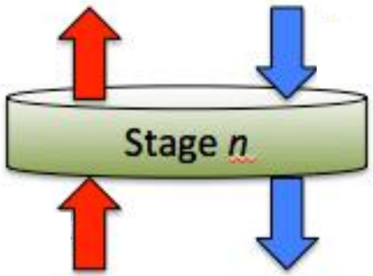


What steps to develop a model for analysis or design?

(1)Overall material balance
(2)Component balance.
(3)Then single equilibrium stage.
(4)Then the stages together.
(5)Possibly energy balance.

Fig. 1.16; Seader, Henley, and Roper, *Separation Process Principles,* 3rd Ed.

Sketch a stage; label; write balances.



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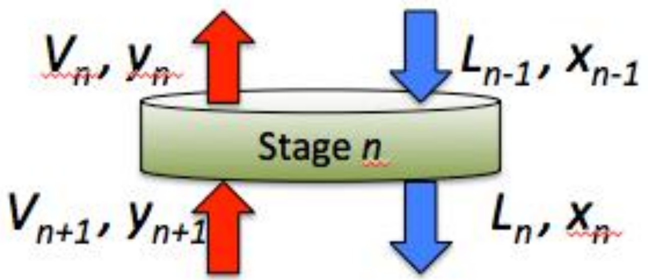

- Key point: Streams L_n and V_n are equilibrated.*
- We have to find the stage equilibrium compositions for all species, plus T and P.

*Well, maybe not COMPLETELY equilibrated. Mass transfer issues?